Bell States, Density Matrices, Partial Trace, and Measurements

1) The four Bell states (two qubits)

Let $\{|0\rangle, |1\rangle\}$ be the computational basis for a single qubit. For two qubits, the computational basis is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

The **Bell states** are the four maximally entangled two-qubit states

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right), \qquad \left|\Phi^-\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right), \tag{1}$$

$$\left|\Psi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right), \qquad \left|\Psi^-\right\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right). \tag{2}$$

They form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ and are eigenstates of correlated observables (e.g. parity and certain two-qubit Pauli products).

2) Density matrix for all four Bell states

For a **pure** state $|\psi\rangle$, the density operator is

$$\rho \;=\; |\psi\rangle \langle\psi| \,.$$

Example:

$$\rho_{\Phi^+} = \left|\Phi^+\middle\rangle\middle\langle\Phi^+\right| = \frac{1}{2}\Big(|00\rangle \langle 00| + |00\rangle \langle 11| + |11\rangle \langle 00| + |11\rangle \langle 11| \Big).$$

Similarly, expanding each Bell projector:

$$\rho_{\Phi^-} = \frac{1}{2}\Big(|00\rangle \langle 00| - |00\rangle \langle 11| - |11\rangle \langle 00| + |11\rangle \langle 11| \Big), \tag{3}$$

$$\rho_{\Psi^+} = \frac{1}{2}\Big(|01\rangle \langle 01| + |01\rangle \langle 10| + |10\rangle \langle 01| + |10\rangle \langle 10| \Big), \tag{4}$$

$$\rho_{\Psi^-} = \frac{1}{2}\Big(|01\rangle \langle 01| - |01\rangle \langle 10| - |10\rangle \langle 01| + |10\rangle \langle 10| \Big). \tag{5}$$

2) Matrix forms in the computational basis

Use ordering $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Then

$$\rho_{\Phi^+} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad \rho_{\Phi^-} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

$$\rho_{\Psi^+} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \rho_{\Psi^-} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

3) Proof that the Bell states form an orthonormal basis

To show they form an orthonormal basis it suffices to prove:

(i) Normalization: $\langle \Phi^{\pm} | \Phi^{\pm} \rangle = 1$ and $\langle \Psi^{\pm} | \Psi^{\pm} \rangle = 1$. For example,

$$\langle \Phi^{+} | \Phi^{+} \rangle = \frac{1}{2}\big(\langle 00|00 \rangle + \langle 00|11 \rangle + \langle 11|00 \rangle + \langle 11|11 \rangle \big) = \frac{1}{2}(1 + 0 + 0 + 1) = 1.$$

(ii) Orthogonality: inner products between distinct Bell states vanish. Example:

$$\langle \Phi^{+} | \Phi^{-} \rangle = \frac{1}{2}\big(\langle 00|00 \rangle - \langle 00|11 \rangle + \langle 11|00 \rangle - \langle 11|11 \rangle \big) = \frac{1}{2}(1 - 0 + 0 - 1) = 0.$$

Similarly, $\langle \Phi^{\pm} | \Psi^{\pm} \rangle = 0$ because $|00\rangle, |11\rangle$ are orthogonal to $|01\rangle, |10\rangle$.

(iii) Completeness: There are 4 orthonormal vectors in a 4D space $\Rightarrow$ they form a basis.

6) Pure and mixed states (definitions)

Density operator: a quantum state can be represented by a positive semidefinite operator ρ with $\mathrm{Tr}(\rho) = 1$.

Pure state:

$$\rho \text{ is pure} \iff \rho = |\psi\rangle\langle\psi| \iff \rho^2 = \rho \iff \mathrm{Tr}\left(\rho^2\right) = 1.$$

Mixed state: a probabilistic ensemble of pure states $\{\,(p_i, |\psi_i\rangle)\,\}$,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \qquad p_i \geq 0, \quad \sum_i p_i = 1.$$

Then typically $\rho^2 \neq \rho$ and

$$\mathrm{Tr}\left(\rho^2\right) < 1.$$

Example: maximally mixed single-qubit state $\rho = \frac{I_2}{2}$ has $\mathrm{Tr}\left(\rho^2\right) = \frac{1}{2}$.

4) Reduced density matrix via partial trace (full details)

Let ρ_{AB} be a two-qubit density matrix on $\mathcal{H}_A \otimes \mathcal{H}_B$. The reduced state of qubit A is

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \sum_{b\in\{0,1\}} (I_A \otimes \langle b|)\, \rho_{AB}\, (I_A \otimes |b\rangle)\,.$$

Worked example for $|\Phi^+\rangle$:

$$\rho_{\Phi^+} = \frac{1}{2}\Big(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11| \Big).$$

Take Tr_B term-by-term, using $\mathrm{Tr}_B\left(|ab\rangle\langle cd|\right) = \langle d|b\rangle\, |a\rangle\langle c|$:

$$\mathrm{Tr}_B(|00\rangle\langle 00|) = \langle 0|0\rangle\, |0\rangle\langle 0| = |0\rangle\langle 0|\,, \tag{6}$$

$$\mathrm{Tr}_B(|00\rangle\langle 11|) = \langle 1|0\rangle\, |0\rangle\langle 1| = 0, \tag{7}$$

$$\mathrm{Tr}_B(|11\rangle\langle 00|) = \langle 0|1\rangle\, |1\rangle\langle 0| = 0, \tag{8}$$

$$\mathrm{Tr}_B(|11\rangle\langle 11|) = \langle 1|1\rangle\, |1\rangle\langle 1| = |1\rangle\langle 1|\,. \tag{9}$$

Therefore,

5) Measurement on one qubit (projective measurement)

Consider measuring qubit A in the computational basis $\{|0\rangle, |1\rangle\}$.

Projectors on the two-qubit space:

$$P_0 = (|0\rangle\langle 0|) \otimes I, \qquad P_1 = (|1\rangle\langle 1|) \otimes I.$$

For a pre-measurement state ρ_{AB},

$$p(0) = \mathrm{Tr}(P_0 \rho_{AB}), \qquad p(1) = \mathrm{Tr}(P_1 \rho_{AB}), \qquad p(0) + p(1) = 1.$$

Post-measurement (collapsed) state conditioned on outcome $k \in \{0, 1\}$:

$$\rho_{AB}^{(k)} = \frac{P_k \rho_{AB} P_k}{p(k)}.$$

Example: for $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, one finds

$$p(0) = \frac{1}{2}, \quad p(1) = \frac{1}{2},$$

and

$$\rho_{AB}^{(0)} = |00\rangle\langle 00|, \qquad \rho_{AB}^{(1)} = |11\rangle\langle 11|.$$

7) Pure-Python code: partial trace + von Neumann entropy

Goal: compute $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ and

$$S(\rho_A) = -\,\mathrm{Tr}(\rho_A \log_2 \rho_A) = -\sum_{i=1}^{2} \lambda_i \log_2 \lambda_i,$$

where $\{\lambda_i\}$ are eigenvalues of the 2×2 matrix ρ_A.

No NumPy used: we implement matrix ops with Python lists and compute eigenvalues analytically.

7) Code (standard Python libraries only)

```
import math
import cmath

def outer(v):
    """Outer product |v><v| for a statevector v (length 4)."""
    n = len(v)
    return [[v[i] * v[j].conjugate() for j in range(n)] for i in range(n)]

def partial_trace_B(rho):
    """
    Partial trace over qubit B for a 2-qubit density matrix rho (4x4).
    Basis ordering: |00>,|01>,|10>,|11>.
    Returns rho_A (2x2).
    """
    rhoA = [[0j, 0j],[0j, 0j]]
    # Indices: |a b> mapped to idx = 2*a + b
    for a in (0,1):
        for a in (0,1):
```

8) Pure-Python code: measurement on one qubit

We implement projective measurement of qubit A in $\{|0\rangle, |1\rangle\}$.

For a pure statevector $|\psi\rangle$, outcome probabilities are

$$p(k) = \langle\psi| P_k |\psi\rangle, \quad P_k = (|k\rangle\langle k|) \otimes I,$$

and the post-measurement state is

$$|\psi_k\rangle = \frac{P_k |\psi\rangle}{\sqrt{p(k)}}.$$

Below: compute $p(0), p(1)$ and the conditional post-measurement statevectors.

8) Code (standard Python libraries only)

```
import math

def measure_qubit_A_statevector(psi, outcome):
    """
    Measure qubit A in computational basis on a 2-qubit statevector psi (len=4).
    outcome = 0 or 1.
    Returns (p, psi_post) where psi_post is normalized statevector.
    """
    # Projector keeps basis states with A=outcome:
    # indices 2*outcome + b for b in {0,1}
    keep = [2*outcome + 0, 2*outcome + 1]
    p = sum((psi[i].conjugate() * psi[i]).real for i in keep)
    if p <= 0.0:
        return 0.0, [0j,0j,0j,0j]
    norm = 1.0 / math.sqrt(p)
    psi_post = [0j,0j,0j,0j]
    for i in keep:
        psi_post[i] = psi[i] * norm
```

8) Qiskit code: Bell state + measurement on IBM hardware

We create a Bell state, measure one qubit, and run on IBM Quantum using Qiskit Runtime `SamplerV2`. The same circuit also works on simulators.

Notes:

- The circuit prepares $|\Phi^+\rangle$: apply H on qubit 0, then $\mathrm{CNOT}(0 \to 1)$.
- Measure only qubit 0 (the “one-qubit measurement” request).
- On real hardware, you need an IBM Quantum API token.

8) Qiskit Runtime (SamplerV2) example

```
from qiskit import QuantumCircuit
from qiskit.transpiler.preset_passmanagers import generate_preset_pass_manager

from qiskit_ibm_runtime import QiskitRuntimeService, SamplerV2 as Sampler

# 1) Authenticate (set your token once, e.g. via environment variable)
# service = QiskitRuntimeService(channel="ibm_quantum", token="YOUR_TOKEN")
service = QiskitRuntimeService() # uses saved account configuration

# 2) Choose a backend (you can also specify a particular device name)
backend = service.least_busy(operational=True, simulator=False)

# 3) Build circuit: create Bell state |Phi+> and measure qubit 0 only
qc = QuantumCircuit(2, 1)
qc.h(0)
qc.cx(0, 1)
qc.measure(0, 0)
```

Quick checks / expected results

For $|\Phi^+\rangle$ and measurement of qubit A in the computational basis:

- $p(0) = p(1) = 1/2$.
- Conditional post-measurement state is $|00\rangle$ or $|11\rangle$.
- Reduced state $\rho_A = I_2/2$ for any Bell state $\Rightarrow$

$$S(\rho_A) = 1 \text{ bit.}$$

These statements can be verified by the pure-Python scripts and by running the Qiskit circuit.